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Annexure-D

~~Advisory III~~

ADVISORY 0N USAGE OF ARTIFICIAL INTELLIGENCE (AI) MODELS (ONLINE MODELS AND SMART DEVICES WITH INBUILT DEEPSEEK FEATURE)

1. **Background.** Artificial Intelligence (AI) denotes the simulation of human cognitive functions in machines enabling them to think, learnt and make decisions. This domain encompasses a diverse range of technologies, including machine learning, natural language processing, computer vision, and robotics. However, AI technologies necessitates careful consideration of data privacy and ethical implications.

2. **Aim**. The advisory is to sensitise all IN personnel regarding implications of Artificial Intelligence (AI) models and smartphones/ smart devices with prebuilt AI models in terms of cybersecurity and national security.

3. Commonly used AI models include conversational models, contextual search, image generation, speech recognition, video generation, gaming, simulation, semi-autonomous, content generation, data analysis and forecasting models. However, they are subject to inherent risks of exposing personal data, propagation of biased perspectives, lack of transparency; surveillance, technical risks associated with system integrity and data storage within extraterritorial jurisdiction. Some common AI models are mentioned below:

Ser	Model	Application Area
a)	OpenAI GPT-4	Chatbot used for content generation, contextual search and creative writing
b)	OpenAI DALLE	image generation, creative design
c)	Google BERT	Contextual search, Natural Language Processing (NLP)
d)	DeepSeek R1	Chatbot, content generation and reasoning

4. **Online/ Cloud Hosted AI Models.** A query is initiated by the user which functions as a prompt by the model. This query is sent to the AI server where it is processed, biased as per its own filters and response is sent to the user. The underlying data processing conducted without explicit knowledge of the user, involves the following: -

   (a) The AI model may keep a repository of previous queries, anticipated response formats, user preferences and other customisations. This information is utilised to refine and modulate future responses. While this feature was designed to enhance user experience, this data, if subjected to unauthorised access by malicious actors or mandated disclosure to government authorities, presents a potential vulnerability for individual profiling, intelligence gathering and predictive analysis of future actions.

   (b) The AI servers of the popular models are situated outside India thereby having limited or no data sovereignty.

   (c) Responses generated by AI models are subject to modulation and potential bias, reflecting the national policies and strategic objectives of the jurisdictions in which they originate.

   (d) With the growing integration of AI powered features in the smartphones, smart devices and home appliances and devices equipped with inbuilt AI models present potential security, privacy, and operational risks, particularly for individuals handling sensitive and classified information.

5. **Risks.** Common AI platforms have inherent risks of data collection, user input storage, keystroke pattern recording, undefined data retention periods, data sharing with third-party entities, potential access by respective governmental authorities, international and jurisdictional issues, lack of granular consent mechanism, potential data breaches, unilateral policy modifications, automated data collection like IP address, device ID, keystroke patterns, etc. Devices with integrated AI models may possess the capability to remotely access devices for updates or diagnostics, which could create security vulnerabilities if exploited by malicious actor. This access might enable unauthorised users to control devices or monitor use activities without consent.

6. **Risk Scenarios w.r.t Military Personnel.** In recent years, a proliferation of AI platforms has emerged within the public domain. AI platforms such as DeepSeek which have a direct affiliation with China, engage in collection of substantial personal and network information pertaining to users. Sample scenarios and risks associated with their usage are highlighted below. These factors necessitate careful consideration when evaluating the utilisation of such applications and models for personal or organisational purposes.

Ser	Scenario	Impact
a)	**Location and Deployment Tracking.** AI models such as DeepSeek logs IP addresses, device identifiers and behavioural patterns. Adversaries monitoring such data could potentially identify login recurring activity originating from specific geographic coordinates.	(i) If adversarial intelligence services could successfully correlate these access points as military locations, they may deduce patterns of troop deployments. (ii) Strategic locations could be mapped and targeted for cyber-attacks, surveillance, or even kinetic attacks. (iii) The microphone, and AI-driven sensors activated remotely to classified discussions. camera, may be capture
b)	**Data Breach Leading to Social Engineering Attacks.** Anti-national elements may compromise servers of such AI models and exfiltrate chat histories, device-specific data, and record user interactions. Thereby analyse stored of military conversations personnel discussing operational matters.	(i)Adversaries use extracted data to craft highly convincing phishing emails spear phishing attacks on military personnel. (ii)They impersonate trusted contacts using leaked chat histories to gain access to classified networks or extract further intelligence. (iii) Psychological profiling of individuals could be used for blackmail, coercion, or insider by recruitment adversarial intelligence agencies.
c)	**Compromised AI Outputs Leading to Disinformation.** Anti-national elements may infiltrate such AI models and manipulate its responses. Military personnel relying on these models for open source intelligence, translations, or quick research may receive altered misleading information	(i)Manipulated outputs from AI models may be leveraged to execute disinformation campaigns, resulting in operations disruptions. (ii)Adversaries may disseminate fabricated narratives designed to influence strategic decision-making.

7. **Recommendations.** Threats, risks and vulnerabilities in use of AI models, raise serious security and privacy concerns. Therefore, following is recommended: -

a) Ensure AI model comply with Data Privacy laws - DPDP Act, GDPR etc.
b) Use only for non-sensitive tasks.

c) Apply data masking and sanitisation before submitting to AI model
d) Monitor for Data Exfiltration Risks.
e) Beware of AI-Generated Phishing & Deepfakes.
f) Verify AI Output for Bias & Misinformation.
g) Restrict API Access to AI models.
h) Modify Firewall blocklist to block / grant selective access using IAM.
i) Prepare a local policy for use of AI models based on ICSP Guidelines.
j) Educate users on AI risks, cybersecurity, and misinformation.
k) Anonymise prompts/data inputs.
l) Use disposable accounts to hide identity.
m) Clear AI chat histories and cache.
n) Disable browser autofill for AI sites.
o) Understand that AI responses can be biased based on their country of origin or intent.
p) Enforce Zero Trust Architecture before deploying AI models.
q) Disable all optional features.
r) Minimise data sharing.
s) Disable GPS when not in use, turn off location services for excessive apps, and use VPN to mask IP location.
t) Use end-to-end encrypted messaging apps.
u) Regularly review and adjust app permissions to ensure that applications only have access to necessary data only.
v) Users should consider turning off features like facial recognition and predictive text to minimise data collection and potential exploitation.
w) Activate full-device encryption to protect stored data from unauthorised access in case of loss or theft.

9. **Conclusion.** End users should be sensitised that AI Models have very limited security features to protect data privacy and AI models in smart devices possess inherent potential security and privacy risks. Hence, AI models are **NOT to be used for official work** and great caution be exercised when used for personal requirements. Cybersecurity is not an individual task; it's a shared responsibility', every member of the Defence community must be cautious of the potential vulnerabilities associated with such AI models.